Prabhavit Gupta

LLM Application Engineer | Multimodal Extraction, Gemini, Evals & GCP Pipelines

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Executive Summary

- **Production LLM Pipelines & Structured Extraction:** AI Application Engineer with 3 years commercial software experience designing, shipping, and hardening multimodal extraction systems that turn messy real-world inputs (PDFs, HTML, emails, images) into validated, schema-compliant product database records.
- **Multimodal Workflows (Google Gemini, Anthropic, OpenAI):** Deep expertise building production workflows with Google Gemini (multimodal document/image understanding, structured JSON schema outputs, temperature tuning, and cost/latency tradeoffs), alongside Claude and GPT models. Creator of open-source document parsing architectures with deterministic validation gates.
- **Evaluation Discipline, Cost Control & Reliability:** Proven track record establishing continuous evaluation harnesses (golden sets, regression test suites, RAGAS, DeepEval) that prevent quality drift across prompt and model updates. Experienced optimizing token expenditures, caching layers, and asynchronous job queues on GCP / Cloud Run.

Technical Skills & Core Competencies

- **LLMs & Multimodal Extraction:** Google Gemini (Multimodal text/vision, structured outputs), Anthropic Claude, OpenAI SDK, Pydantic Schema Enforcement, Instructor, Outlines, Context Pruning.
- **Evaluation & Observability:** RAGAS Benchmarking, DeepEval, Golden Dataset Regression Suites, Langfuse Distributed Tracing, Latency & Token Budgeting, Failure Mode Triaging.
- **Backend & Cloud Infrastructure:** Python (AsyncIO, FastAPI, Pydantic, PyTest), GCP (Cloud Run, Cloud Functions, Firestore), Docker, PostgreSQL (pgvector), Redis Semantic Caching.
- **Document & Data Pipelines:** PDF/HTML/Email Parsing, OCR Fallbacks, Deduplication, Taxonomy Mapping, Idempotent Rescans, Automated CI/CD Testing with GitHub Actions.
- **Client & Studio Delivery:** Agency/Consultancy Multi-Client Delivery, Extreme Ownership, C1+ English, US EST Working Hours Overlap (through 5 PM EST).

Professional Experience

LLM Application Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Global)

- Architected and hardened end-to-end production LLM extraction pipelines for European and US client platforms, processing high volumes of unstructured PDFs, invoices, and email interactions.
- Leveraged Google Gemini and Claude multimodal capabilities to parse complex document layouts and images, producing structured Pydantic schemas stored directly into client product databases.
- Established automated evaluation harnesses using golden benchmark datasets, testing precision, field completeness, and duplicate detection across model updates.
- Implemented semantic caching mechanisms with Redis and token tiering strategies, cutting API operational expenses by 35% and maintaining predictable sub-second response times.
- Instrumenting real-time observability and distributed trace debugging using Langfuse, tracking token burn, latency percentiles, and automated retry fallbacks across long-running async jobs.
- Collaborated closely with product leads and mobile developers to define strict JSON field contracts, validation rules, and automated human-in-the-loop review queues.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Engineered asynchronous microservices in Python (FastAPI) and TypeScript for automated document classification, entity normalization, and database storage.
- Built vector indexing layers using PostgreSQL pgvector and Redis, accelerating search speeds across messy enterprise data records.
- Developed automated unit and integration regression test suites using PyTest, achieving 90%+ test coverage in continuous CI/CD deployment pipelines.
- Managed multi-tenant client configurations, ensuring reliable execution and data privacy across distinct customer environments.

Key Engineering Projects

- **LLM Multimodal Document Parser (github.com/prab-gupta/llm-doc-parser):** Open-source production pipeline extracting structured attributes from multi-format inputs (PDFs, HTML emails, scans) using Gemini and

Claude with schema validation and error recovery.

- **Continuous LLM Regression & Quality Gate (Python + DeepEval + RAGAS):** Automated CI/CD evaluation harness running synthetic test suites to verify extraction completeness, hallucination absence, and token budgets prior to production deployments.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, decision analysis, statistical reasoning, and quantitative problem-solving.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency).
- **Work Authorization & Availability:** EU Permanent Resident (Permesso di Soggiorno UE a Tempo Indeterminato) / Italian Partita IVA ready for independent remote retainer contracts. Available immediately with daily US EST working hours overlap.